

Chapter 4: Learning through mastery

Digital Education (CS-411)

September 29, 2026

From exploration to consolidation

Exploration

- Carefully design the materials
- Comparisons, playing with differences
- Conflict: trapping predictions
- Push reflection

Consolidation



There is a time for telling

Can you figure out what to do?

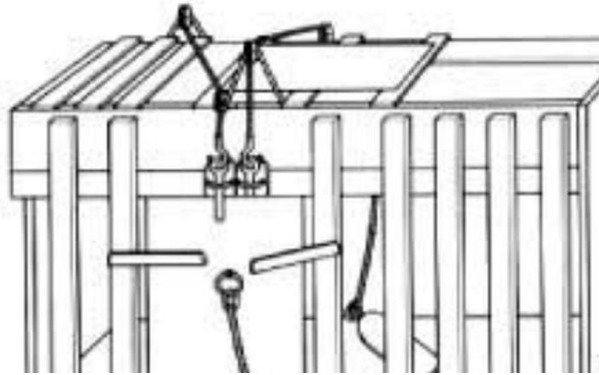
There is something I want you to do.

- You can try anything.
- I will not give you any instructions.
- A clap is the only feedback you will receive.



Edward L. Thorndike (1874–1949) [The Law of Effect](#)

Behavior followed by satisfying consequences becomes more likely to be repeated;
behavior followed by unpleasant consequences becomes less likely.

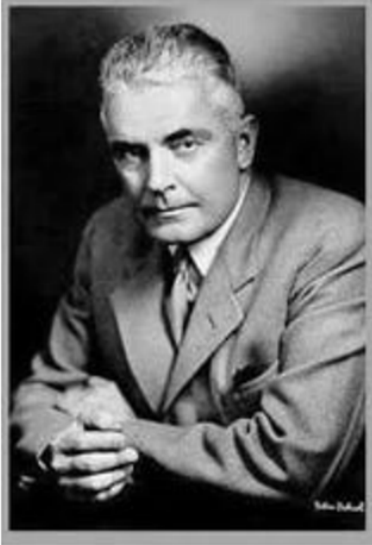


Puzzle box



Escape time decreases across trials

Behaviorism



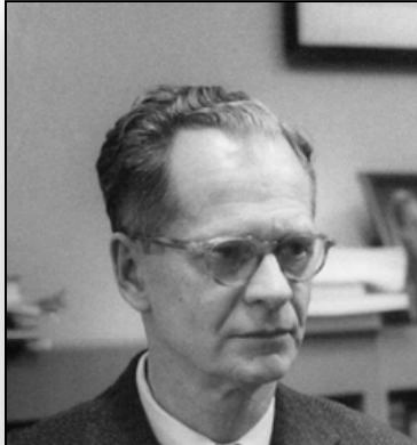
John B. Watson

1878–1958

“ Give me a dozen healthy infants, well-formed, and my own specified world to bring them up in and I'll guarantee to take any one at random and train him to become any type of specialist I might select — doctor, lawyer, artist, merchant-chief and, yes, even beggar-man and thief, regardless of his talents, penchants, tendencies, abilities, vocations, and race of his ancestors.

— John B. Watson, *Behaviorism* (1930)

[The Little Albert Experiment](#)

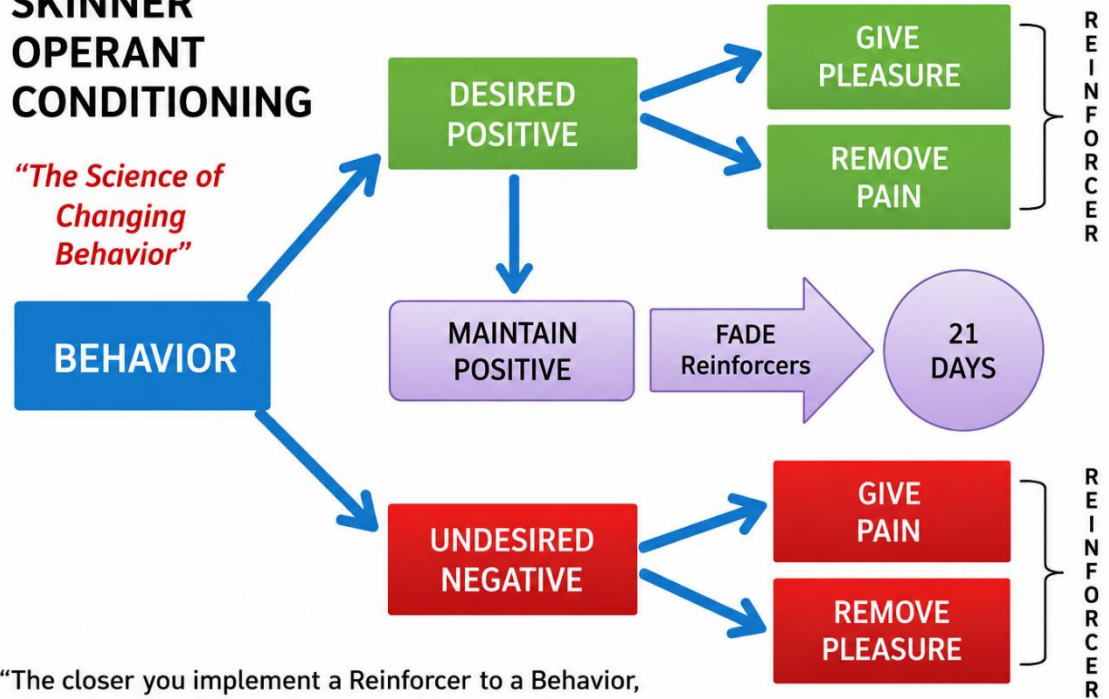


B. F. Skinner (1904–1990)

Operant conditioning

SKINNER OPERANT CONDITIONING

*"The Science of
Changing
Behavior"*



"The closer you implement a Reinforcer to a Behavior,
the greater the chance of changing behavior"

Key ideas of behaviorism

- ① The mind is a black box; the focus is on **observable behavior**
- ② Associations are strengthened through **immediate feedback**
- ③ The learner is continuously **active**
- ④ **Small steps** increase the probability of positive feedback

Constructivist vs. behaviorist approach

Constructivist approach	Behaviorist approach
Errors can be productive	Aim for frequent success
Delayed telling to allow discovery	Immediate feedback
Larger, more open problem space	Small, carefully sequenced steps
Learner explores alternatives	Learner repeatedly responds
Difficulty can provoke knowledge construction	Difficulty is managed to maintain successful responding

Reward

Core learning mechanism:

Rewarding a behavior makes it more likely to be repeated; rewarding successive approximations can shape new behavior.

Why it works:

Rewards can motivate learners to repeat rewarded behaviors in similar situations. By reinforcing progressively closer approximations, a complex target behavior can be built up gradually.

Example:

A math program gives students practice problems. After several correct answers in a row, the learner unlocks new items for customizing their character.

Can you determine the day of the week?

20 May 2041

No calendar. No phone. No computer.

Doomsday algorithm



John Conway, 1973

- The Gregorian calendar moves in cycles of 400 years
- Each year has a certain weekday (*doomsday*) on which easy to remember dates fall
- Century anchor: the doomsday at the start of the century

Doomsday dates: 4/4, 6/6, 8/8, 10/10, 12/12, 5/9, 9/5, 7/11, 11/7

Century anchors: 1800 – Friday, 1900 Wednesday, 2000 Tuesday

Find Doomsday of year

$$41 = 3 \cdot 12 + 5$$

$$[5/4] = 1$$

$$3 + 5 + 1 = 9$$

$$9 \bmod 7 = 2$$

$$\text{Tuesday} + 2 = \text{Thursday}$$

Find nearby doomsday date
and count from there

May 9 is a Thursday

$$20 - 9 = 11$$

$$11 \bmod 7 = 4$$

Thursday + 4 = **Monday**

Your turn

- What weekday was 20 July 1969?
- What weekday was 2 September 1888?

Worked examples

Core learning mechanism:

Learners study a step-by-step demonstration of how to complete a procedural task.

Why it works:

Worked examples let novices observe well-defined solution steps and expert thinking. They reduce unnecessary search so learners can focus on the steps and relationships that produce a solution.

Example:

To learn the Doomsday algorithm, students first observe a complete solution in which the instructor demonstrates and explains each step of the method.

Deliberate Practice

Core learning mechanism:

Learners engage in focused and effortful practice to develop specific skills and concepts beyond their current abilities.

Why it works:

Deliberate practice improves accuracy and fluency and makes skills less effortful to execute. This frees cognitive resources for more complex tasks and helps learners recognize meaningful patterns.

Example:

Practicing Doomsday problems with different century anchors and counting directions.

Feedback

Core learning mechanism:

Feedback allows learners to sense the discrepancy between what they did and what they should have done, enabling them to adjust future actions.

Why it works:

Learners need information about whether they are on the right track. Effective feedback identifies the discrepancy and, ideally, points to what should be changed or improved.

Example:

A student solves $0,4 \cdot 0,3 = 0.12$. The system identifies the incorrect placement of the decimal point and explains how to correct it.

Formative Feedback

Effective feedback answers three questions

Where am I going?

Feed up

What are the goals?

How am I going?

Feed back

What progress is being made?

Where to next?

Feed forward

What should I do next?



Each feedback question can operate at four levels

Task

How well the task is understood or performed

Process

Processes and strategies used to understand or perform the task

Self-regulation

Monitoring, directing, and regulating one's own actions

Self

Personal evaluations and praise about the learner

Feedback

Current Status

[Hide](#)

You are tasked with identifying which function grows fastest as $n \rightarrow \infty$. You attempted to select an exponential function, but your choice was incorrect.

You correctly recognized that exponential functions dominate polynomial ones, but you missed critical comparisons. The key error is not analyzing the base of the exponential ($\log_3(33) > 3$) and ignoring the polynomial terms in other options.

Next Steps

[Hide](#)

Task Changes

Compare all exponential bases directly (e.g., $\log_3(33) > 3$).

Convert all functions to the same base (e.g., express $n^{3\log_3 n}$ as $3^{3(\log_3 n)^2}$).

Analyze the exponents' growth rates (e.g., $(\log n)^2$ vs. n).

Recheck the constant factors' impact on asymptotic growth.

Strategy Tips

Always convert functions to the same base when comparing growth rates.

Use logarithms to simplify exponents and compare polynomial vs. exponential growth.

Sanitize your answer by testing small n values to verify intuition.

Self-Regulation

Plan to break down complex expressions into simpler components before comparing.

Monitor your assumptions by asking, "Does this hold for all $n > N$?"

Reflect on why initial guesses (like 3^n) might be incorrect.

Set a goal to systematically compare all options before finalizing answers.

Your recognition of exponential growth is a strong start—keep refining your comparisons of bases and exponents!

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Skinner's Teaching Machine

Student response
window

Question/
answer window

Question /
answer disk

Answer strip

Score /
Advance Lever

Instructional **disks** containing questions and answers are placed inside the machine. Students:

1. **Read** a question through the window
2. **Write** their response on a paper strip
3. **Compare** their response with the correct answer
4. **Move a lever** to record whether their answer was correct
5. **Advance** to the next question

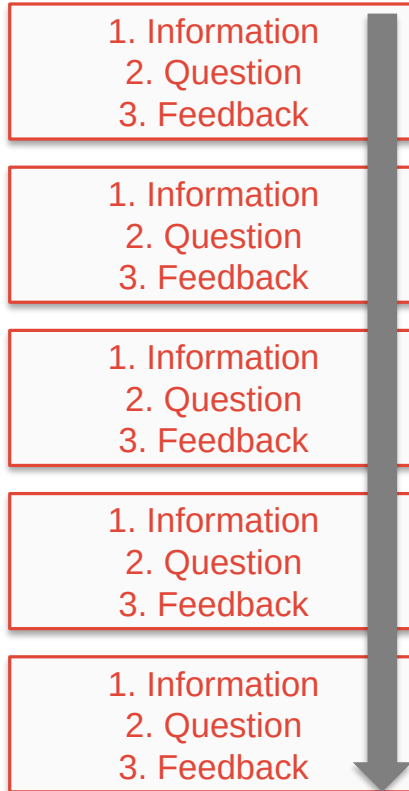
The machine keeps score and controls progression.

Table 2. PART OF A PROGRAM IN HIGH-SCHOOL PHYSICS

The machine presents one item at a time. The student completes the item and then uncovers the corresponding word or phrase shown at the right.

SENTENCE TO BE COMPLETED	WORD TO BE SUPPLIED
1. The important parts of a flashlight are the battery and the bulb. When we "turn on" a flashlight, we close a switch which connects the battery with the _____.	bulb
2. When we turn on a flashlight, an electric current flows through the fine wire in the _____ and causes it to grow hot.	bulb
3. When the hot wire glows brightly, we say that it gives off or sends out heat and _____.	light
4. The fine wire in the bulb is called a filament. The bulb "lights up" when the filament is heated by the passage of a(n) _____ current.	electric
5. When a weak battery produces little current, the fine wire, or _____, does not get very hot.	filament
6. A filament which is less hot sends out or gives off _____ light.	less
7. "Emit" means "send out." The amount of light sent out, or "emitted," by a filament depends on how _____ the filament is.	hot

Frame-based models / E-Learning



1. **Decomposition:** Segment complex content into a sequence of learning steps, each containing an elementary piece of information.
2. **Keep the student active:** Ask the student to process each new piece of information.
3. **Immediate feedback:** Provide feedback after each response.
4. **Self-paced progression:** Let the student move on at their own speed.

From small steps to prerequisites

Skill1 is prerequisite to *Skill2* if
 $p(\textit{Skill2} \mid \neg \textit{Skill1}) \ll p(\textit{Skill2} \mid \textit{Skill1})$

Skill1: « 6 + 9 »

Skill2: « 26 + 39 »

Skill1: to find the verb

Skill2: to agree it with the subject

Skill1: mean

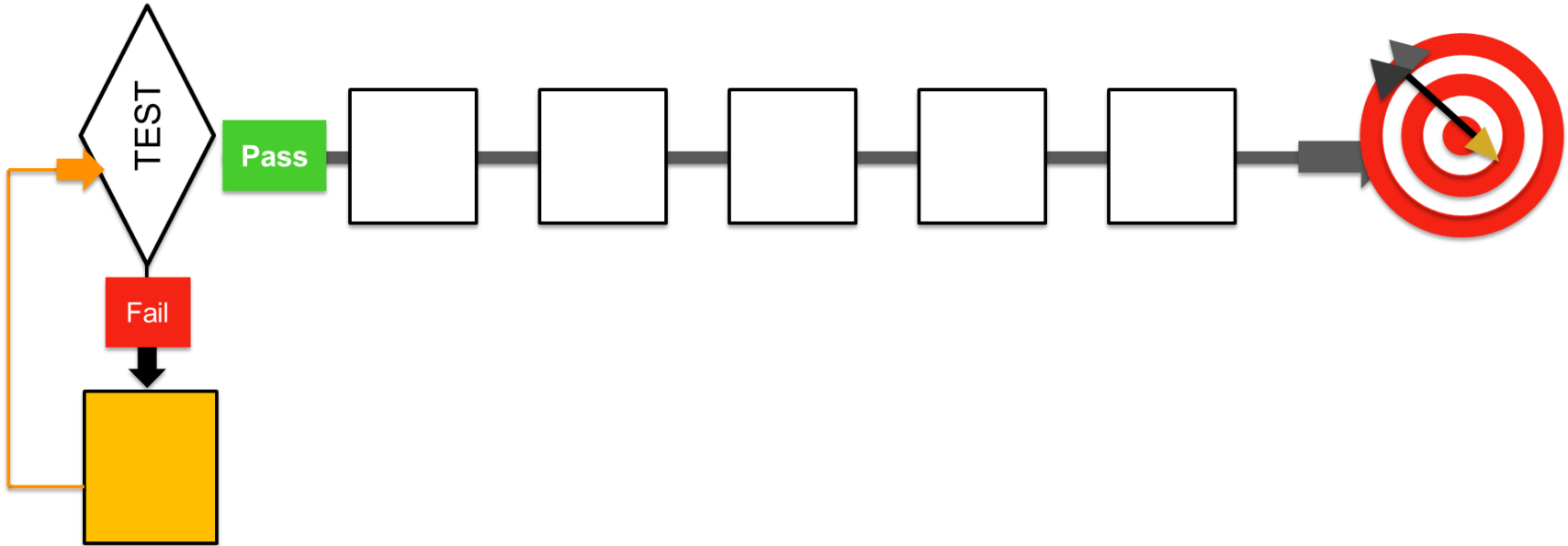
Skill2: standard deviation

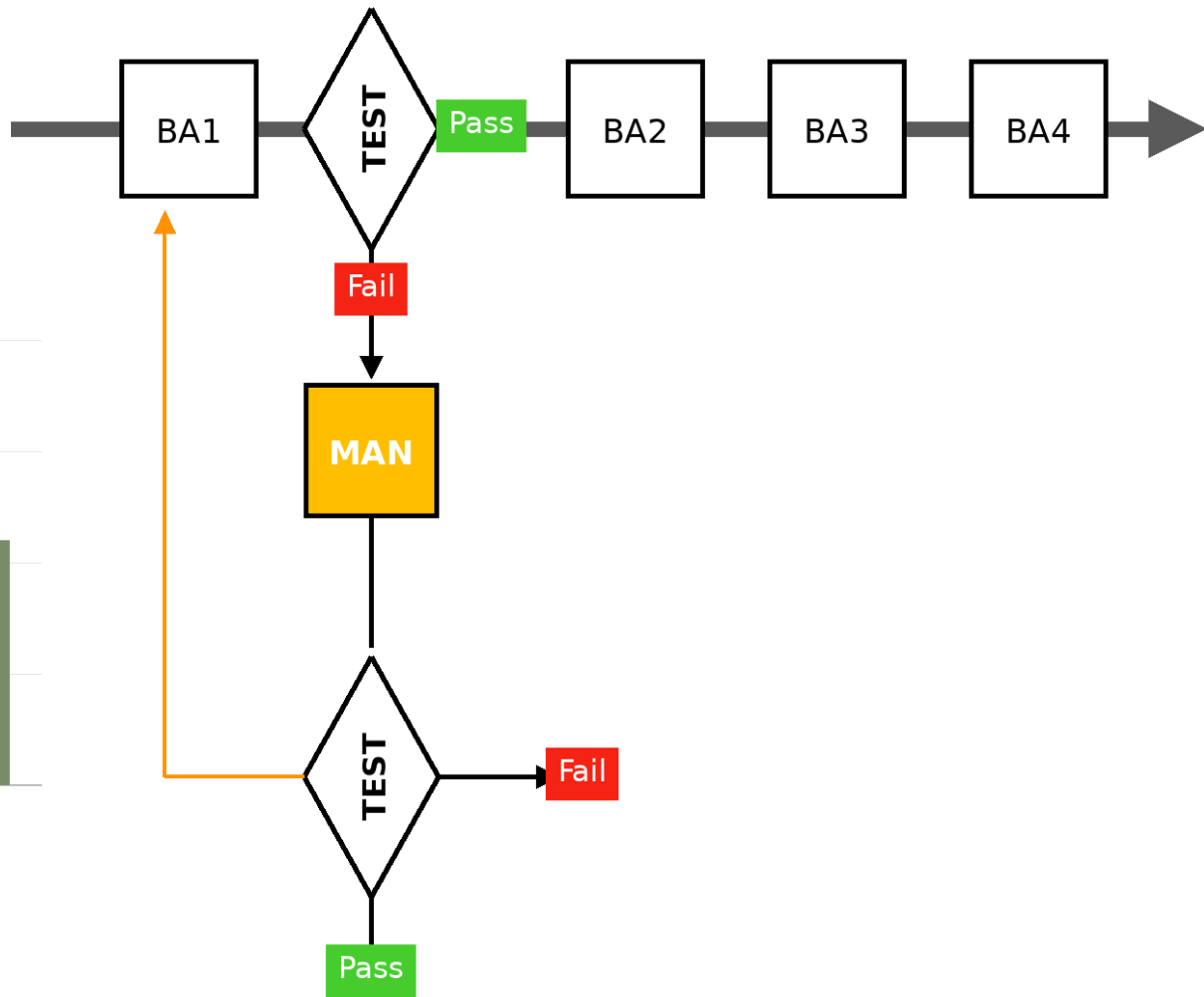
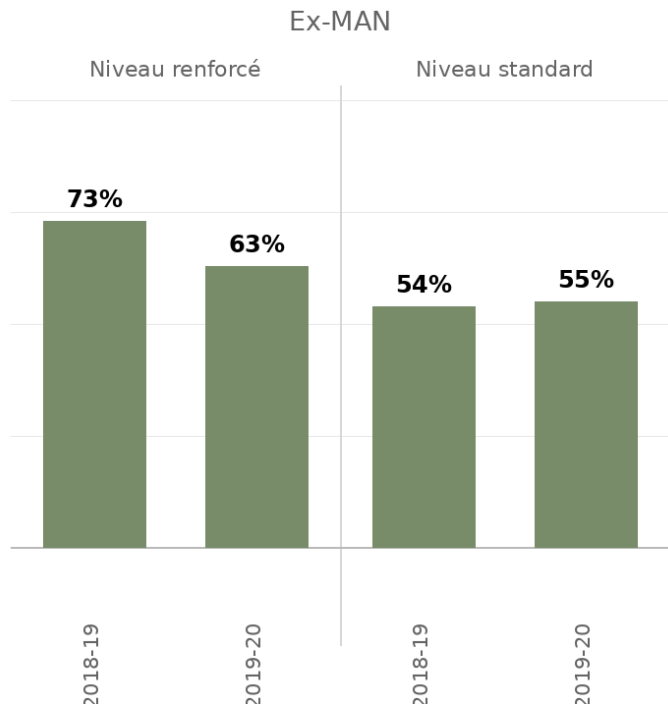
Skill1: to read

Skill2: to code

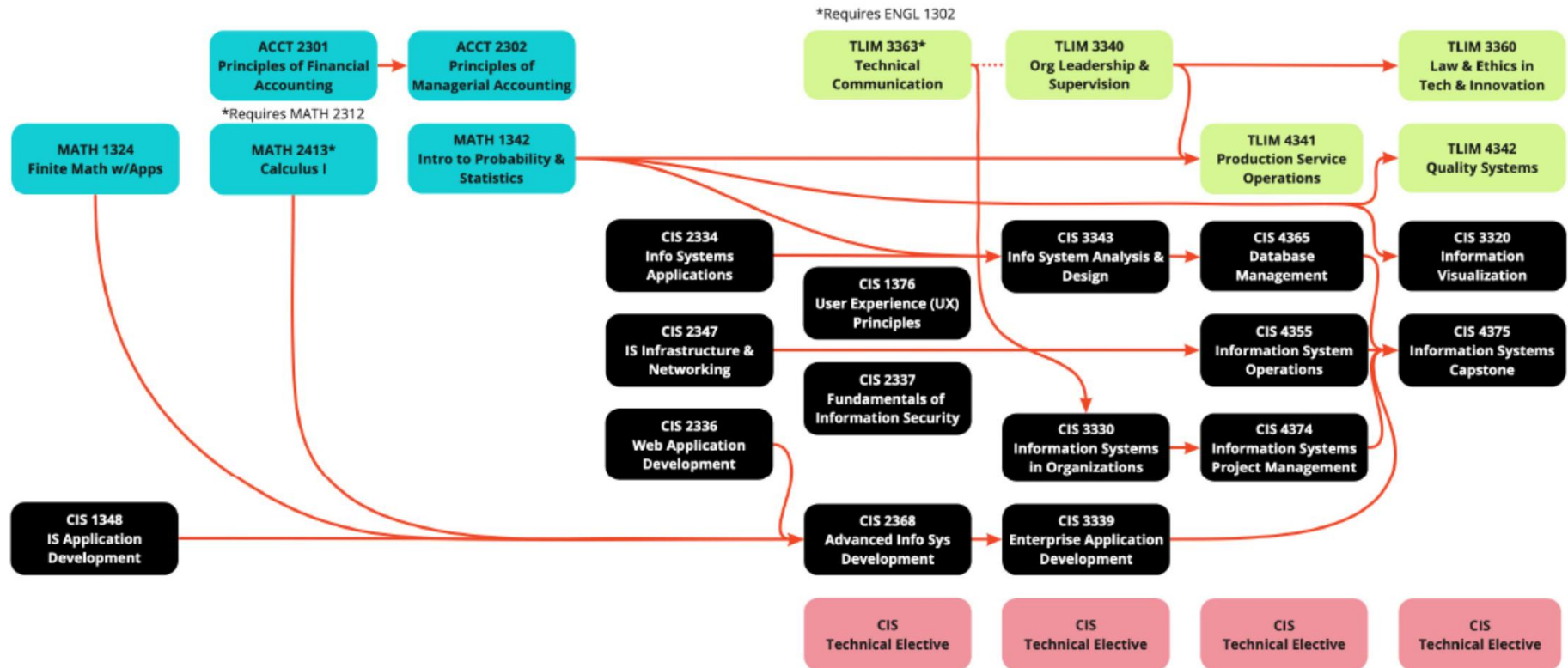
Mastery learning

If prerequisites are missing, remediate them before going on



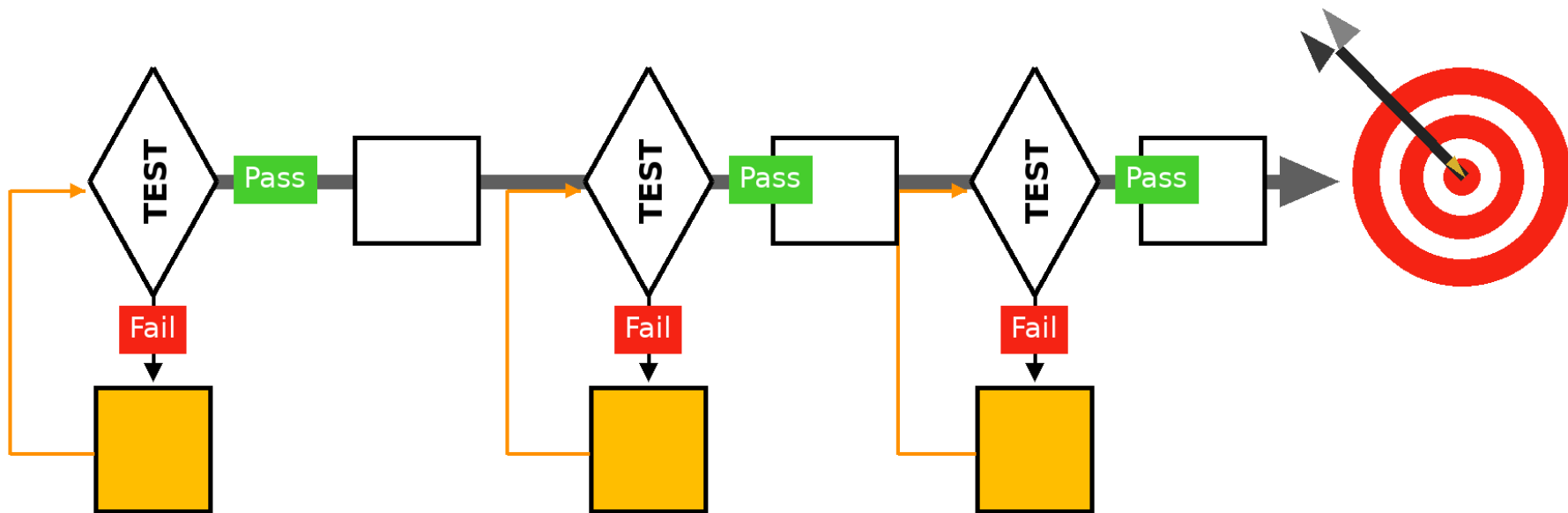


Course prerequisite map

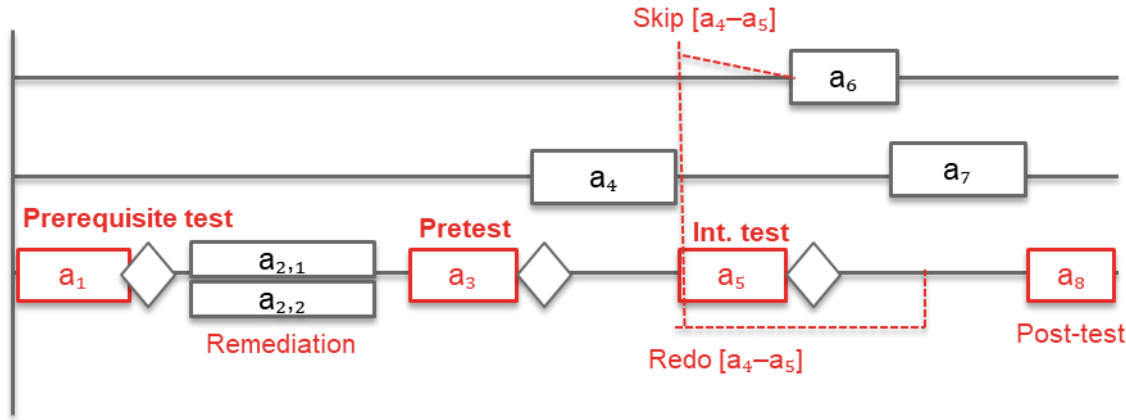


Mastery learning

Control mastery of $skill_i$ before you go to $skill_{i+1}$



Modular Instruction



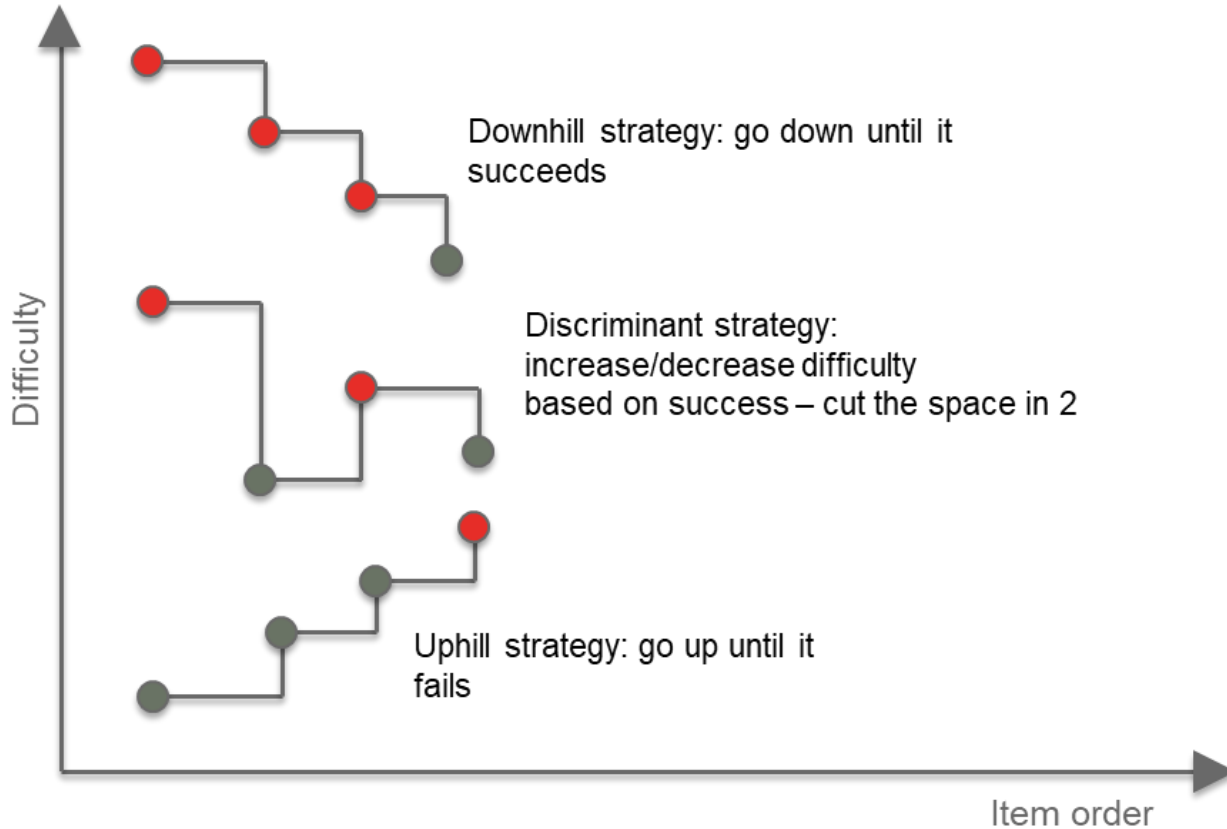
Pre-requisite test: Does the learner has the pre-requisite to start the course ?

Pre-test: Should the learner skip some modules ?

Intermediate-test: Did the learner reach the objectives of this module ?

Post-test: Did the learner reach the objectives of this course?

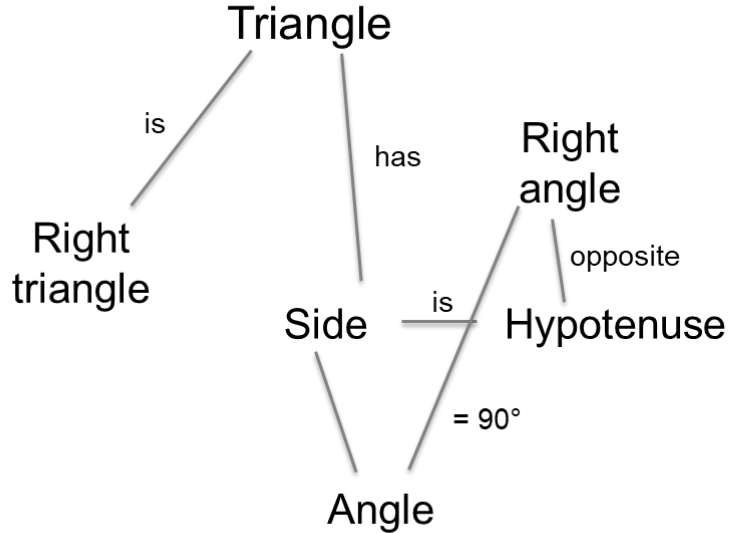
Adaptive Testing



Decomposition

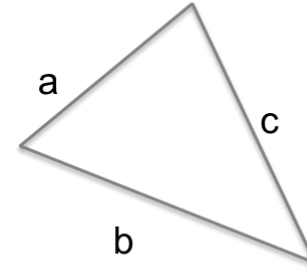
What does the learner need to KNOW?

Knowledge decomposition



What does the learner need to DO?

Skill decomposition



Procedure

1. Measure A
2. Compute A^2
3. Measure B
4. Compute B^2
5. $A^2 + B^2 = C^2$
6. $C = \sqrt{C^2}$

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Learner explores alternatives	Learner repeatedly responds
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Instructional design starts with :

~~What should learners know at the end ?
(which they did not know at the beginning)~~

What should learners be able to do at the end ?
(which they could not do at the beginning)

How will I know they are able to do it ?

What would be the exam questions or tasks ?

Instructional design works backwards :



What are they able to
do at the beginning

Pre-Requisites

What should learners be
able to do at the end ?

(which they could not do
at the beginning)

Objectives

How can we sequence activities?

When is an activity a_i useful for an activity a_{i+1} ?

1. a_i is a pre-requisite for a_{i+1}

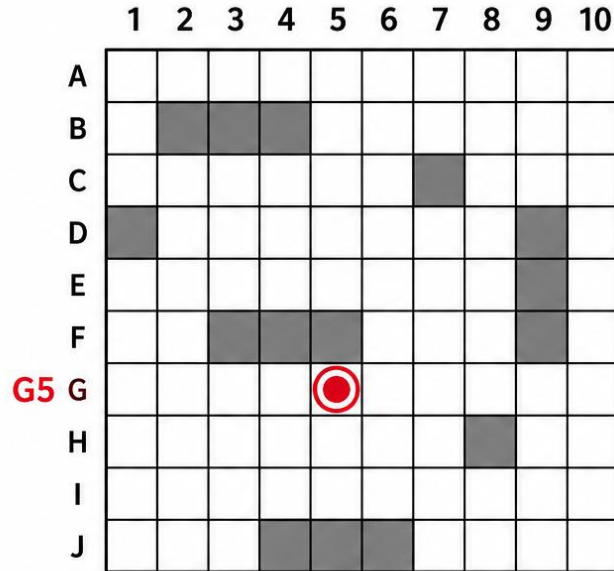
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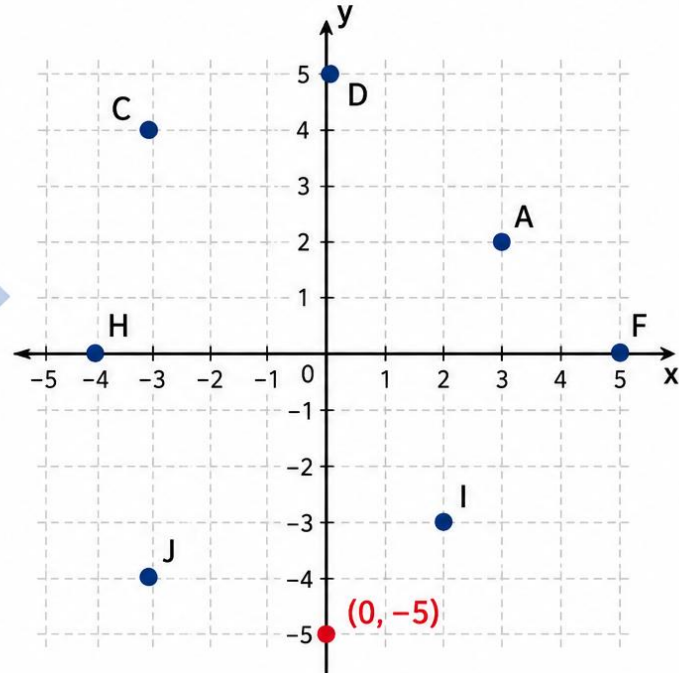
1. a_i is a pre-requisite for a_{i+1}
2. a_i activates a cognitive schema needed in a_{i+1}

Advance organizer

Battleship



Cartesian Coordinates System



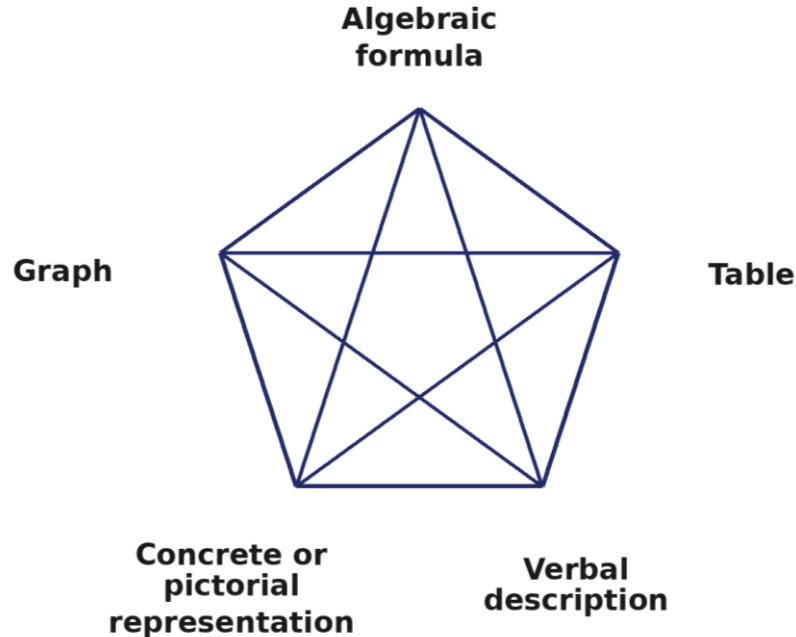
How can we sequence activities?

When is an activity a_i useful for an activity a_{i+1} ?

1. a_i is a pre-requisite for a_{i+1}
2. a_i activates a cognitive schema needed in a_{i+1}
3. a_{i+1} varies the representation used in a_i

Shift representation

a_{i+1} represents content differently from a_i



How can we sequence activities?

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4. ...

Two tasks...

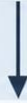
A chocolate bar costs 5 CHF. How many bars can you buy for 25 CHF ?

A man walks 1 km in 5 min. How many km can he walk in 25 min ?

Context A



1 chocolate bar
costs **CHF 5**.



How many chocolate bars
can you buy for **CHF 25**?



Context B



A person walks
1 km in **5 minutes**.



How far can they walk
in **25 minutes**?

**Transferring knowledge between contexts
is **not** (often) spontaneous.**

How can we support transfer?

1. By reinstantiating skills in various contexts
2. By targeting a deeper understanding

Represent 35_{10} in base 3.

Calculate directly in base 3:

$$\begin{array}{r} 122_3 \\ + 222_3 \\ \hline \end{array}$$

From exploration to consolidation

Exploration

- Carefully design the materials
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Consolidation



There is a time for telling

How can we sequence activities?

When is an activity a_i useful for an activity a_{i+1} ?

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2. a_i activates a cognitive schema needed in a_{i+1}
3. a_{i+1} varies the representation used in a_i
4. a_{i+1} practices the transfer of skills acquired in a_i

Instructional design is not about choosing one theory. It is about combining learning mechanisms in ways that serve the learning objectives.

Exploration → Worked Example

Contrasting Cases → Deliberate Practice → Feedback

Worked Example → Practice + Undoing → Consolidation